Francesca Ciccarelli

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Short Bio

Francesca Ciccarelli is a Ph.D. Candidate in Information and Communication Technologies at the University of L'Aquila. She holds a B.Sc. in Mathematics and an M.Sc. in Applied Data Science. Her research interests focus on Machine Learning and Trustworthy Artificial Intelligence, with particular interest in Machine Unlearning. She has also gained research experience in cybersecurity and healthcare applications of Artificial Intelligence.



Research Experience

Nov 2024–	**Ph.D. Candidate in Information and Communication Technologies**, University of L'Aquila, Italy. Fields of study include Data Analysis, Data Mining, Machine Learning, Decision Models and Optimization, and Big & Open Data Management and Processing.
Feb–Jul 2024	**Intern**, Leonardo S.p.A., Italy. Research activities on Key Performance Indicators (KPIs) and Key Risk Indicators (KRIs) in the cybersecurity domain. Activities included data collection, data inspection and exploration, the design of a system of indicators, and analysis of the results.
Oct 2023–Apr 2024	**"Antonio Alfonsi" Fellowship**, I.B.M.A.I., Italy. Application of Machine Learning and Artificial Intelligence technologies in healthcare, including the development of a mobile application using Android Studio and Java.

Education

2024–	**Ph.D.**, Information and Communication Technologies, University of L'Aquila, Italy.
2022–24	**M.Sc.**, Applied Data Science, University of L'Aquila, Italy, 110/110 cum laude. Thesis: *"Strategic Weighing: Methodologies for Cybersecurity Evaluation and Visualization of KPIs and KRIs"*. Development of a methodology for the evaluation of risk analysis in the corporate environment, with application on real data.
2014–22	**B.Sc.**, Mathematics, University of L'Aquila, Italy, 90/110. Thesis: *"Statistical analysis of official COVID-19 data"*. Application of the different phases of Data Science to COVID-19 data in the province of L'Aquila, comparing pre- and post-vaccine data.

Honours & Awards

2024	Awarded a **Travel Grant** for the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24), Barcelona, Spain.
2023	Awarded the **"Antonio Alfonsi" Fellowship**, I.B.M.A.I., Italy.

Teaching

2026	**Tutor, Tutorato Amico Programme**, Department of Information Engineering, Computer Science and Mathematics (DISIM), University of L'Aquila, 150 hours.
2026	**Laboratory Instructor**, Networks, University of L'Aquila, 4 hours.
2025	**Laboratory Instructor**, Data Mining, University of L'Aquila. Laboratory on Frequent Itemset Mining, 2 hours.
2023	**Instructor**, Centro Studi La Meta / Onsemi Group. Instructor for the seminar *"From Topology to Knot Theory"* within the Masterclass 2023 project.

Service

Organization

2026	Co-organizer of Workshop on Web & Graphs, Responsible Intelligence, and Social Media (WEB&GRAPH 2026), WSDM (ACM International Conference on Web Search and Data Mining) 2026.

Reviewing

2026	ACM Web Conference (The Web Conference). *External Reviewer*.
2026	ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases). *External Reviewer*.
2026	ITADATA (Italian Conference on Big Data and Data Science). *External Reviewer*.
2026	WIPE-OUT (Workshop on Innovations, Privacy-preservation, and Evaluations of machine Unlearning Techniques).
2026	WEB&GRAPH (Workshop on Web & Graphs, Responsible Intelligence, and Social Media) @ WSDM 2026.

Volunteering

2024	**Volunteer**, 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24), Barcelona, Spain. Minimum of 10 hours of conference volunteering.

Presentations

Posters

P1. **Ciccarelli, F.**, D'Angelo, A. & Stilo, G. *Towards a Novel Visual Evaluation of Algorithmic Bias: Insights on the Italian Academic System*. Undergraduate Consortium, 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24), Barcelona, Spain, 2024.

Publications

Other Academic Contributions

O1. **Ciccarelli, F.**, D'Angelo, A. & Stilo, G. *Towards a Novel Visual Evaluation of Algorithmic Bias: Insights on the Italian Academic System*. Presented at the Undergraduate Consortium of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '24), 2024.

Selected Projects

Analysis and Visualization of COVID-19 Data: *National and global levels*. Application of the data-mining process to COVID-19 vaccination data from different countries, including preprocessing and analysis of vaccination coverage and daily vaccinations.

Forecasting of People Looking for Employment: *Time series analysis*. Analysis of Italian employment data (1959–2015) and comparison of ARIMA models using ACF/PACF and AIC, BIC, and HQC criteria. The model yielding the lowest RMSE was used for forecasting. Analyses were performed using GretL.

Creation of Web Service for Querying RDF Files in SPARQL: *RDF, SPARQL*. Creation of a web service for querying an open dataset on famous paintings. The project included data cleaning and enrichment with OpenRefine, RDF generation, SPARQL querying with Apache Jena, and REST client/server implementation.

Differential Analysis of microRNA Involved in Breast Cancer: *Biomedical research*. Development of a bioinformatics pipeline for data quality improvement, alignment of miRNA sequences against reference miRNAs and the human genome GRCh38 using Bowtie, and identification of differentially expressed miRNAs using Python.

Application of Machine Learning Algorithms for Binary Classification and Sequence Predictions: *Machine Learning*. Application of KNN and MLP models to breast-cancer classification, development of deep ensemble architectures, comparison of different training strategies, and application of MLPs to sequence prediction.

Skills

Languages

Italian: Native.

English: B2, certified by university exam.

Technical Skills

Programming: Python (NumPy, Pandas, Matplotlib); Java / Android Studio.

Data and tools: GretL, RDF/SPARQL, Apache Jena, OpenRefine, Bowtie, Microsoft Office Suite.

Last updated: October 7, 2026